

Ex.No. 11:

Implementation of IoT-based Smart Agriculture using MATLAB

AIM:

To design and simulate an IoT-based smart agriculture system using MATLAB by monitoring soil moisture levels, controlling water pump status (ON/OFF), and evaluating irrigation efficiency for optimized water resource management.

Objective:

1. To monitor the Soil Moisture (%) level in agricultural fields using IoT-based sensing simulation in MATLAB.
2. To control and display the Water Pump Status (ON/OFF) based on threshold soil moisture conditions.
3. To evaluate the Irrigation Efficiency (%) by analyzing water usage effectiveness and moisture maintenance level.

Objective 1 Matlab Code :

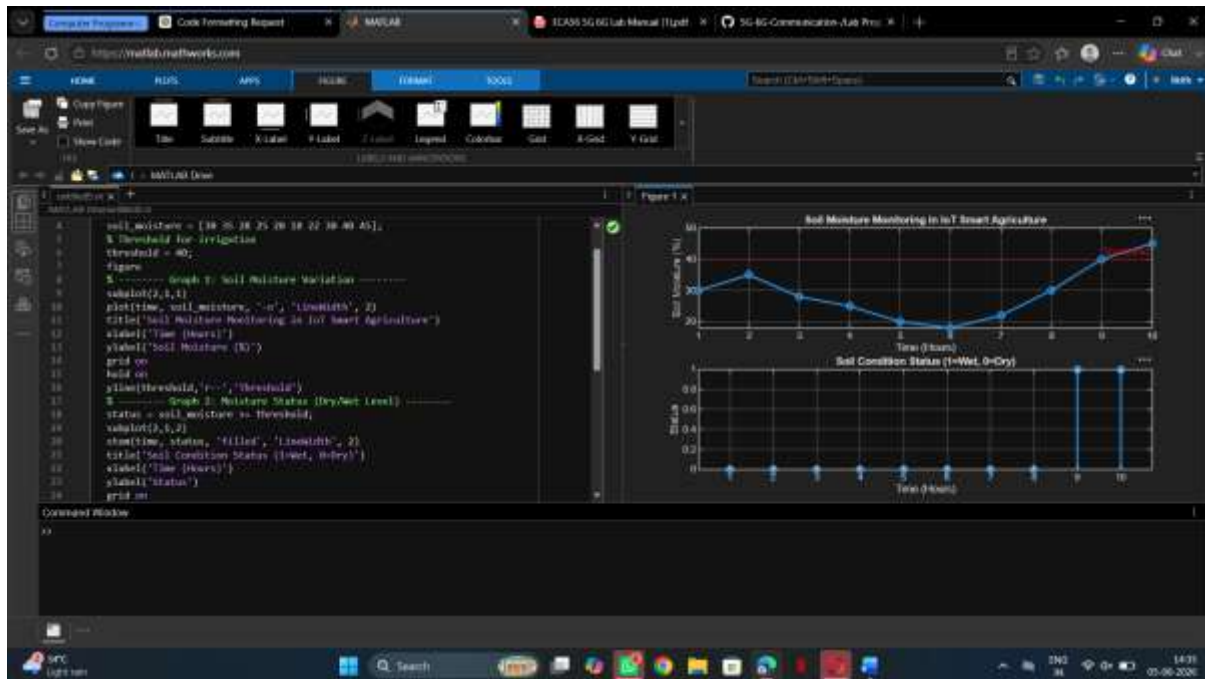
```
clc;
clear;
close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold for irrigation
threshold = 40;
figure;
% ----- Graph 1: Soil Moisture Variation -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Monitoring in IoT Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold, 'r--', 'Threshold')
% ----- Graph 2: Moisture Status (Dry/Wet Level) -----
status = soil_moisture >= threshold;
```

```

subplot(2,1,2)
stem(time, status, 'filled', 'LineWidth', 2)
title('Soil Condition Status (1=Wet, 0=Dry)')
xlabel('Time (Hours)')
ylabel('Status')
grid on

```

OUTPUT :



Objective 2 Matlab Code :

```

clc;
clear;
close all;

% Simulated soil moisture (%)
time = 1:10;

soil_moisture = [30 35 28 25 20 18 22 30 40 45];

% Threshold level
threshold = 40;

% Pump control logic (1 = ON, 0 = OFF)
pump_status = soil_moisture < threshold;

figure;

% ----- Graph 1: Soil Moisture vs Pump Status -----

```

```

subplot(2,1,1)

plot(time, soil_moisture, '-o', 'LineWidth', 2)

title('Soil Moisture vs Pump Control')

xlabel('Time (Hours)')

ylabel('Soil Moisture (%)')

grid on

hold on

yline(threshold, 'r--', 'Threshold')

% ----- Graph 2: Pump Status (ON/OFF) -----

subplot(2,1,2)

stem(time, pump_status, 'filled', 'LineWidth', 2)

title('Water Pump Status (1=ON, 0=OFF)')

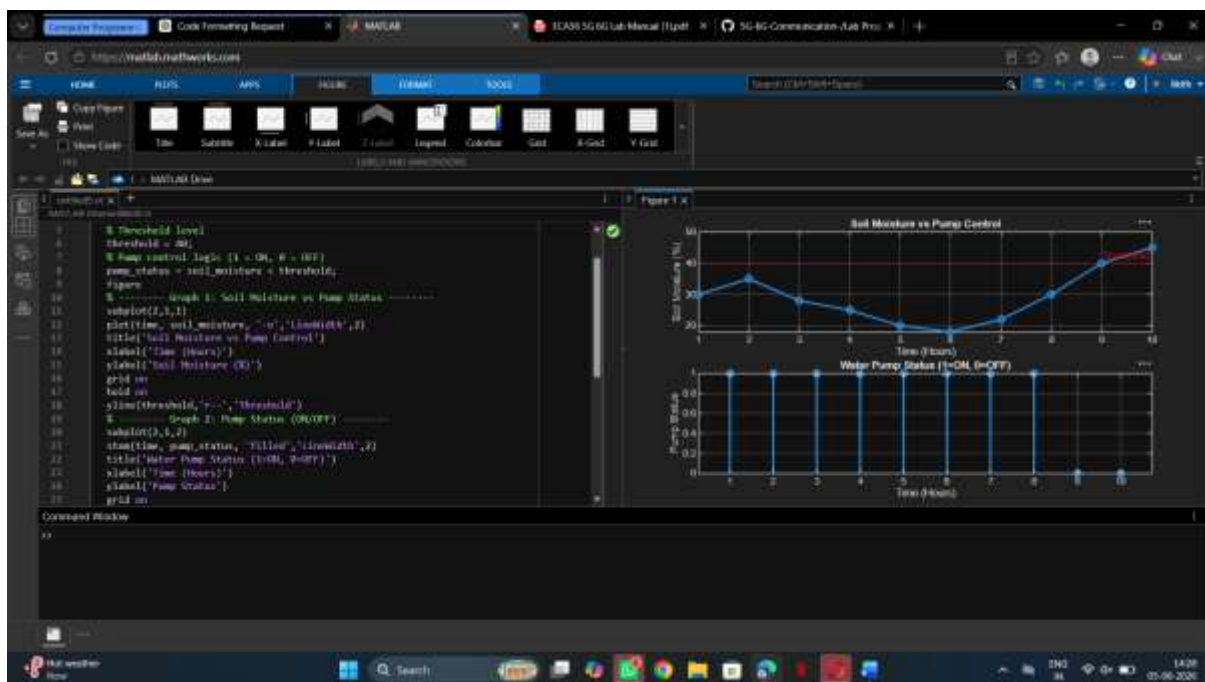
xlabel('Time (Hours)')

ylabel('Pump Status')

grid on

```

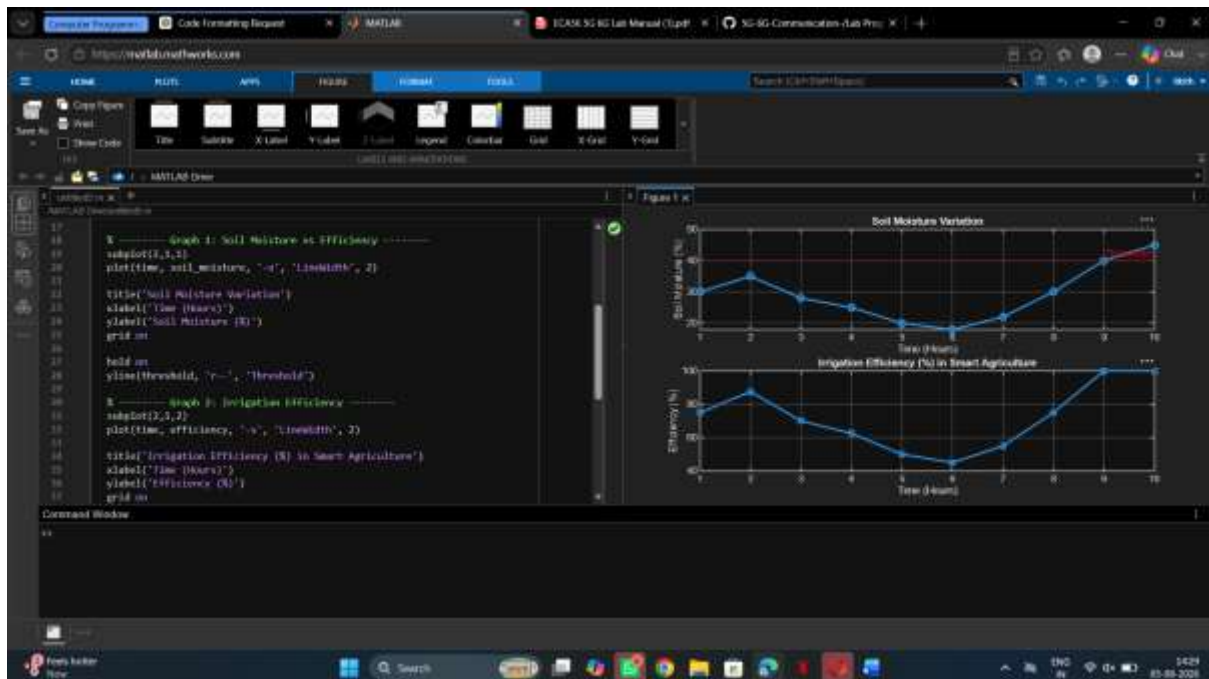
OUTPUT :



Objective 3 Matlab Code :

```
clc;
clear;
close all;
% Simulated soil moisture data (%)
time = 1:10;
soil_moisture = [30 35 28 25 20 18 22 30 40 45];
% Threshold level
threshold = 40;
% Irrigation efficiency calculation (%)
efficiency = (soil_moisture ./ threshold) * 100;
efficiency(efficiency > 100) = 100; % limit to 100%
figure;
% ----- Graph 1: Soil Moisture vs Efficiency -----
subplot(2,1,1)
plot(time, soil_moisture, '-o', 'LineWidth', 2)
title('Soil Moisture Variation')
xlabel('Time (Hours)')
ylabel('Soil Moisture (%)')
grid on
hold on
yline(threshold, 'r--', 'Threshold')
% ----- Graph 2: Irrigation Efficiency -----
subplot(2,1,2)
plot(time, efficiency, '-s', 'LineWidth', 2)
title('Irrigation Efficiency (%) in Smart Agriculture')
xlabel('Time (Hours)')
ylabel('Efficiency (%)')
grid on
```

OUTPUT :



Result:

1. The soil moisture levels were successfully monitored using MATLAB simulation, showing variation over time and identifying dry and wet conditions based on threshold values.
2. The water pump control system was effectively implemented, where the pump automatically switched ON when soil moisture dropped below the threshold and remained OFF when moisture was sufficient.
3. The irrigation efficiency was successfully calculated and analyzed, showing that the system optimizes water usage by maintaining soil moisture close to the desired level.